Understanding Failure Modes in Pedestrian Trajectory Prediction via Interpretable Risk Analysis

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Abstract

Trajectory prediction models are increasingly deployed in safety-critical systems such as autonomous robots and assistive navigation platforms. However, the input conditions under which these models fail remain poorly understood, limiting their safe deployment in real-world environments. This paper presents a model-agnostic framework for systematic failure mode analysis of pedestrian trajectory predictors. The framework combines input-space sensitivity analysis, spatial risk mapping, and interpretable decision tree classifiers to identify and explain prediction failures without requiring access to model internals or training data. I evaluate the framework on a Social Force Model applied to two real-world pedestrian scenes from the ETH dataset, comprising 719 trajectories across distinct environments. Our results show that initial position and orientation are the dominant factors determining prediction reliability, while walking speed plays a minor role. Importantly, failure patterns differ across scenes, suggesting that scene-specific safety monitoring is necessary. The extracted decision tree rules provide human-readable safety conditions that can be directly integrated into runtime monitoring systems, enabling safer robot navigation in human-shared environments.

1 Introduction

Mobile robots operating alongside humans increasingly rely on trajectory prediction models to anticipate pedestrian motion and plan collision-free paths [1]. Applications range from autonomous wheelchairs and service robots to self-driving vehicles in urban environments [2]. In all these settings, prediction failures can lead to unsafe robot behaviour, including collisions or overly conservative manoeuvres that reduce usability.

Despite significant progress in learning-based trajectory predictors, including recurrent neural networks, generative models, and graph-based architectures [3, 4], their deployment in safety-critical contexts remains problematic. A core challenge is that standard evaluation metrics—Average Displacement Error (ADE) and Final Displacement Error (FDE)—provide aggregate performance summaries but do not reveal *when* or *why* a model fails for a specific pedestrian under specific conditions.

This limitation is particularly important for assistive robotics, where the cost of prediction failure is high. A wheelchair navigation system that cannot reliably predict pedestrian motion near doorways or corridor intersections may endanger its user. Understanding the input conditions that trigger prediction failures is therefore a prerequisite for safe deployment.

Existing work on safe trajectory prediction has focused primarily on improving model architectures [5] or providing probabilistic uncertainty estimates [6]. However, relatively little

attention has been paid to *post-hoc failure characterisation*—systematically identifying which input conditions cause an already-trained model to underperform.

In this paper, I address this gap by proposing a model-agnostic failure mode analysis framework with the following contributions:

- A systematic evaluation pipeline that assesses trajectory predictor reliability across a wide range of input conditions without modifying the underlying model.
- Spatial and input-space risk maps that visualise failure-prone regions, providing intuitive safety diagnostics.
- An interpretable decision tree layer that translates empirical risk maps into explicit, human-readable failure rules suitable for runtime monitoring.
- A cross-scene evaluation demonstrating that failure patterns are scene-dependent, motivating the need for scene-specific safety assessment.

I demonstrate the framework on a Social Force Model [10] evaluated on the ETH pedestrian dataset [11], covering two distinct outdoor scenes with a total of 719 pedestrian trajectories.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the proposed framework. Section 4 presents the experimental evaluation. Section 5 discusses the findings and their implications, and Section 6 concludes the paper.

2 Related Work

2.1 Trajectory Prediction

Human trajectory prediction has been studied extensively, with methods ranging from physics-based models to deep learning approaches. The Social Force Model [10] remains a widely used baseline, modelling pedestrian dynamics as a system of attractive and repulsive forces. Data-driven approaches, including LSTMs [3], GANs [4], and transformer-based models [5], have significantly improved prediction accuracy on standard benchmarks. However, high average accuracy does not guarantee reliable behaviour across all input conditions, particularly in edge cases relevant to safety.

2.2 Safety in Robot Navigation

Safe robot navigation in human-shared environments has been approached through chance-constrained planning [7], reachability analysis, and control barrier functions [8]. These methods typically assume that a reliable prediction model is available. The question of how to characterise and mitigate prediction model failures before they propagate to the planner has received less attention.

2.3 Interpretable Machine Learning for Safety

Interpretable ML methods, including decision trees [9] and rule extraction techniques, have been applied to explain model behaviour in healthcare and autonomous driving. Their application to trajectory prediction reliability assessment is relatively unexplored. Our work builds on this direction by using decision trees not to replace the predictor, but to explain its failure conditions in human-readable terms.

2.4 Failure Mode Analysis

Failure mode analysis in the context of neural networks has been studied through adversarial examples, out-of-distribution detection, and stress testing. For trajectory prediction specifically, recent work [1] has surveyed the field but noted the lack of systematic frameworks for characterising failure conditions. Our work directly addresses this gap.

3 Methodology

3.1 Problem Formulation

Let a pedestrian trajectory be defined by a sequence of positions $\mathbf{p}_t = (x_t, y_t) \in \mathbb{R}^2$ over discrete time steps $t \in \{1, \ldots, T\}$. A trajectory predictor f_θ maps an observed sequence of length T_{obs} to a predicted future trajectory:

$$\hat{\mathbf{P}} = f_\theta(\mathbf{p}_1, \ldots, \mathbf{p}_{T_{\text{obs}}}) \tag{1}$$

where $\hat{\mathbf{P}} = (\hat{\mathbf{p}}_{T_{\text{obs}}+1}, \ldots, \hat{\mathbf{p}}_T)$ denotes the predicted future positions. The framework treats f_θ as a black box and analyses its outputs without accessing model internals or training data.

3.2 Evaluation Metrics

Prediction quality is quantified using two standard metrics. The Average Displacement Error (ADE) measures mean positional error over the prediction horizon:

$$\text{ADE} = \frac{1}{T - T_{\text{obs}}} \sum_{t=T_{\text{obs}}+1}^{T} \|\hat{\mathbf{p}}_t - \mathbf{p}_t\|_2 \tag{2}$$

The Final Displacement Error (FDE) measures error at the final predicted position:

$$\text{FDE} = \|\hat{\mathbf{p}}_T - \mathbf{p}_T\|_2 \tag{3}$$

A trajectory is classified as a *failure* if its ADE exceeds a data-driven threshold τ, defined as the 75th percentile of the ADE distribution over the evaluation set:

$$\text{failure}(i) = \mathbb{1}[\text{ADE}_i > \tau] \tag{4}$$

This threshold-based definition provides a principled separation between typical and degraded prediction performance without requiring manual tuning.

3.3 Input Feature Extraction

For each pedestrian trajectory, I extract four scalar features that characterise the initial state at the moment of prediction:

- $s_0 = \|\mathbf{v}_0\|_2$: initial walking speed
- $\phi_0 = \text{atan2}(v_y^0, v_x^0)$: initial orientation in radians, $\phi_0 \in [-\pi, \pi]$
- x_0, y_0: initial position coordinates in metres

These features are chosen because they fully characterise the observable state of the pedestrian at prediction time and are available in any trajectory prediction scenario.

3.4 Risk Mapping

To visualise failure-prone regions of the input space, I define the empirical failure probability over an input bin $\mathcal{U}_{\text{bin}}$ as:

$$R(\mathcal{U}_{\text{bin}}) = \frac{1}{N_{\text{bin}}} \sum_{j=1}^{N_{\text{bin}}} \mathbb{1}\left[\text{ADE}_j > \tau\right] \tag{5}$$

where N_{bin} is the number of trajectories whose initial features fall within the bin. Risk maps are constructed in both the orientation–speed space and the spatial (x_0, y_0) space, providing complementary views of prediction reliability.

3.5 Interpretable Failure Rules

To translate empirical risk maps into actionable safety rules, a decision tree classifier $\mathcal{T}$ is trained on the input features (s_0, ϕ_0, x_0, y_0) using the binary failure labels defined above. The tree is constrained to a maximum depth of 4 and a minimum of 10 samples per leaf, balancing expressiveness with interpretability.

The resulting classifier produces explicit rules of the form:

$$\text{if } y_0 \leq \theta_1 \text{ and } s_0 > \theta_2 \;\Rightarrow\; \text{Failure (prob.} = p) \tag{6}$$

These rules can be directly integrated into a runtime safety monitor: before executing a predicted trajectory, the monitor evaluates the current pedestrian state against the failure rules and triggers a conservative fallback behaviour if a high-risk condition is identified.

3.6 Social Force Model

The Social Force Model (SFM) [10] is a physics-based trajectory predictor that models pedestrian motion as a system of forces:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i^{\text{goal}} + \sum_{j \neq i} \mathbf{f}_{ij} + \mathbf{f}_i^{\text{env}} \tag{7}$$

where $\mathbf{f}_i^{\text{goal}}$ drives agent i toward its goal at preferred speed v_0^i, $\mathbf{f}_{ij}$ represents repulsive interaction with other agent j, and $\mathbf{f}_i^{\text{env}}$ models repulsion from environmental obstacles. In our implementation, I use a constant-velocity baseline (setting interaction forces to zero) to isolate the effect of initial conditions on prediction error, independent of crowd interaction effects.

4 Experiments

4.1 Dataset

I evaluate on two scenes from the ETH pedestrian dataset [11]:

- **ETH scene**: an open outdoor area on a university campus, with 353 valid trajectories after filtering sequences shorter than 5 observations.
- **Hotel scene**: an outdoor area near a hotel entrance with more constrained motion patterns, with 366 valid trajectories.

Table 1: Prediction Performance Across Both Scenes

Metric	ETH	Hotel
Total trajectories	353	366
Mean ADE (m)	0.326	0.238
Mean FDE (m)	0.653	0.480
Failure threshold τ (m)	0.41	0.33
Failure rate	24.9%	25.1%
Decision tree accuracy	79.3%	82.0%

Both scenes provide real-world pedestrian trajectories recorded at 2.5 Hz from a bird's-eye-view camera. The two scenes differ significantly in spatial layout and crowd dynamics, enabling cross-scene generalisation analysis. For each trajectory, the first half is used as the observation window (T_{obs}) and the second half as the prediction target.

4.2 Quantitative Results

Table 1 summarises prediction performance across both scenes.

The Hotel scene shows lower mean ADE (0.238 m vs 0.326 m), suggesting more predictable motion patterns, likely due to the more constrained spatial layout near the hotel entrance. Despite different absolute error levels, both scenes exhibit a comparable failure rate of approximately 25%, confirming that the 75th percentile threshold captures a consistent proportion of degraded predictions.

4.3 Risk Maps

Figure 1 shows the risk maps and comparative analysis for both scenes. In the ETH scene (top left), failures are distributed across all orientation values, with no clear speed-dependent pattern. In the Hotel scene (top right), failures are more concentrated around specific orientation clusters, particularly at orientations corresponding to diagonal movement directions. The ADE distributions (bottom left) confirm that the Hotel scene has lower and more concentrated errors, while the ETH scene exhibits a longer tail of high-error cases.

4.4 Decision Tree Analysis

Figure 2 shows the decision tree learned for the ETH scene. The tree achieves 79.3% accuracy with a maximum depth of 4 and identifies y_0 (initial y-position) as the primary splitting criterion, followed by orientation ϕ_0.

4.5 Feature Importance Analysis

Figure 3 reports the ADE distribution and feature importances for the ETH scene. The feature importance comparison between scenes (Fig. 1, bottom right) reveals a key finding: while orientation (ϕ_0) is consistently important across both scenes (ETH: 0.400, Hotel: 0.432), the relative importance of positional features differs substantially.

In the ETH scene, initial y-position dominates (importance = 0.523), while in the Hotel scene, importance is more evenly distributed across orientation (0.432), speed (0.218), and position (0.176). This suggests that failure mode structure is scene-dependent and that a universal set of failure rules may not generalise across environments.

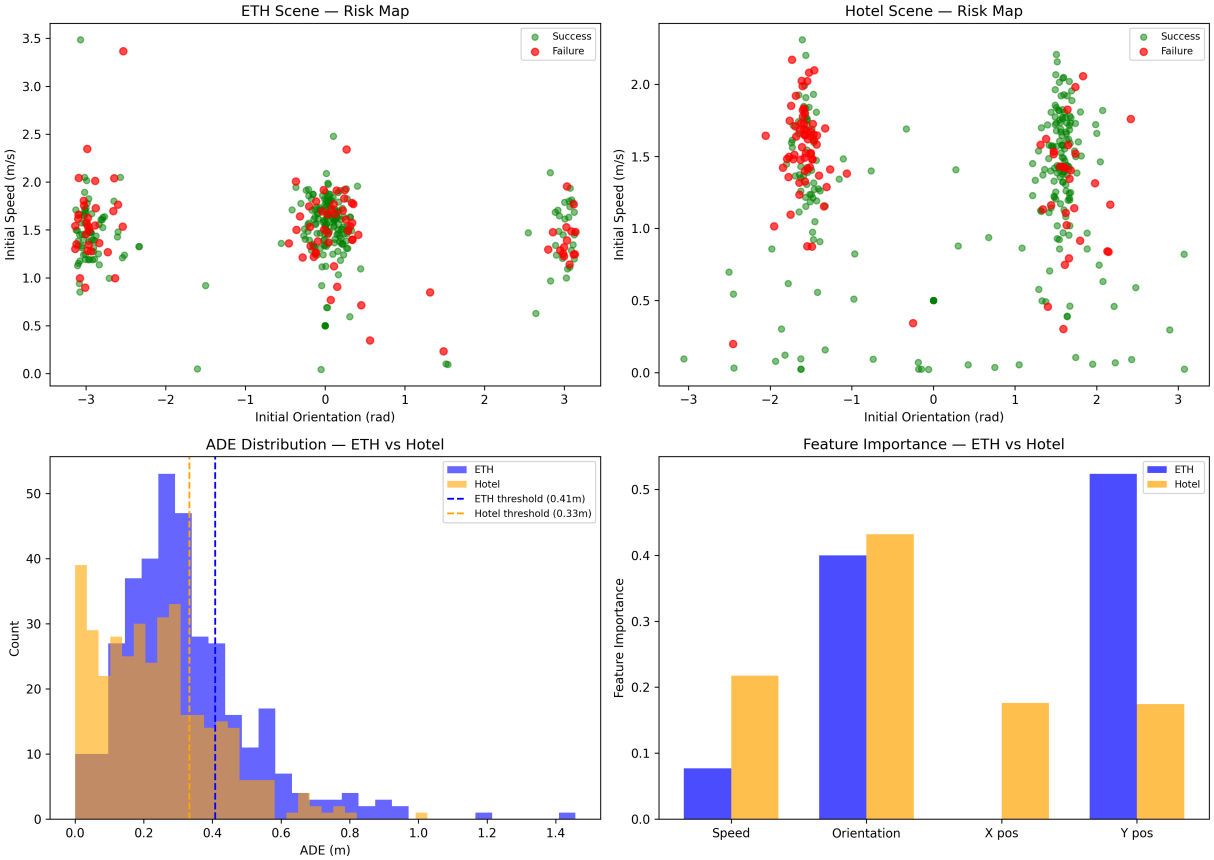


Figure 1: Cross-scene comparison. Top: risk maps in orientation–speed space for ETH (left) and Hotel (right) scenes. Bottom left: ADE distributions. Bottom right: feature importances from decision tree classifiers trained on each scene separately.

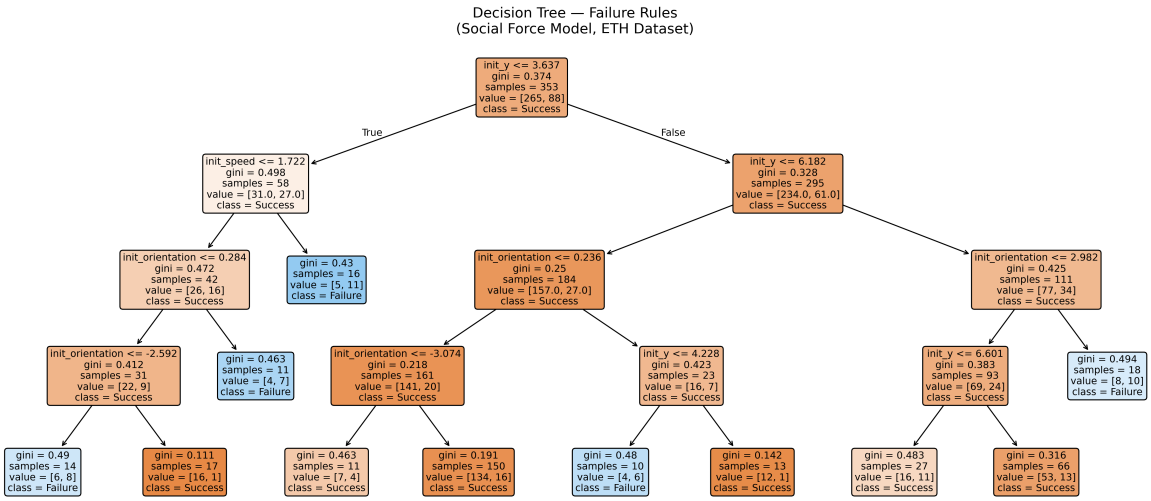


Figure 2: Decision tree for the ETH scene (depth $\leq$ 4, min. 10 samples/leaf). Blue nodes predict Success, orange nodes predict Failure.

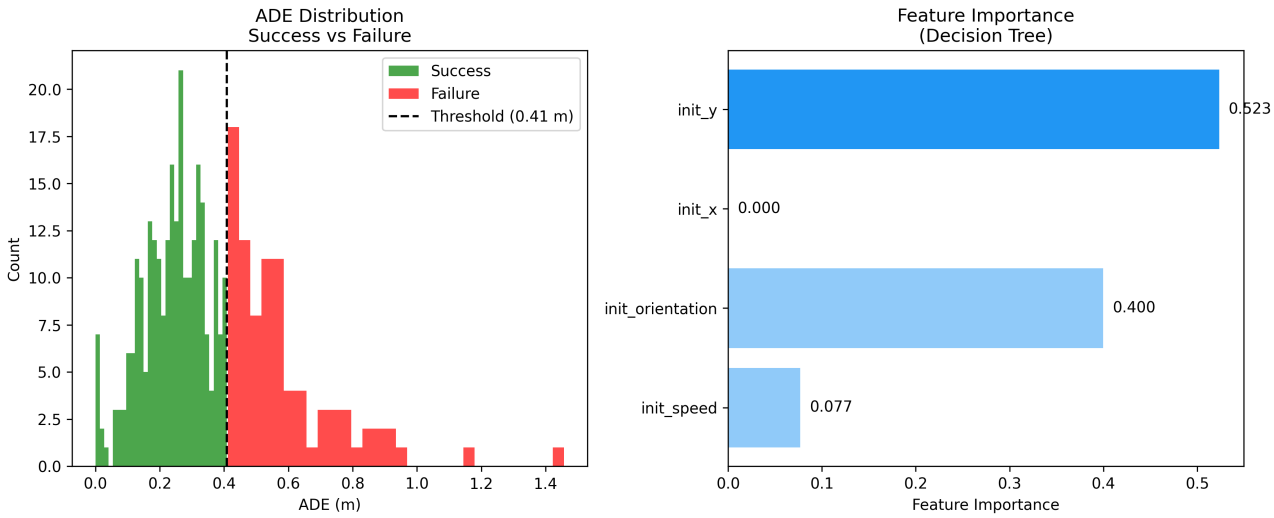


Figure 3: ADE distribution for the ETH scene (left) and decision tree feature importances (right).

5 Discussion

5.1 Interpretation of Findings

The most consistent finding across both scenes is the importance of initial orientation in determining prediction reliability. Pedestrians moving in atypical or diagonal directions relative to the dominant flow of the scene tend to produce higher prediction errors. This is consistent with the known limitation of constant-velocity models, which perform poorly when pedestrians are in the process of changing direction.

The scene-dependent nature of positional importance (dominant in ETH, negligible in Hotel) reflects differences in spatial layout. In the ETH open campus scene, pedestrians near the periphery (low y_0) tend to exhibit more complex motion patterns—turning, stopping, or interacting with others—that are difficult to predict with a constant-velocity baseline. In the more constrained Hotel scene, spatial position is less informative because pedestrians are funnelled through a smaller area with more uniform motion patterns.

5.2 Implications for Safe Robot Navigation

These findings have direct implications for runtime safety monitoring in robot navigation systems. The extracted decision tree rules can be integrated as a pre-execution safety layer: before a robot acts on a trajectory prediction, it evaluates the current pedestrian state against the failure rules. If a high-risk condition is identified—for example, a pedestrian with $y_0 \leq 3.64$ and ϕ_0 outside the safe range—the robot can:

- Reduce its speed to allow more reaction time
- Request a re-prediction with a more conservative model
- Maintain a larger safety margin around the pedestrian
- Alert a human operator in assistive robot contexts

Critically, this safety layer operates without retraining the predictor and adds negligible computational overhead, making it suitable for real-time deployment.

5.3 Limitations

Several limitations should be acknowledged. First, the Social Force Model with constant-velocity baseline is a relatively simple predictor; failure patterns may differ for more expressive neural models. Second, the 75th percentile threshold for failure definition is a design choice that affects failure rates; a lower threshold would capture more conservative safety requirements. Third, the ETH dataset, while widely used, represents specific outdoor environments; generalisation to indoor or more complex scenes requires further evaluation.

5.4 Future Work

Future work will extend the framework to neural trajectory predictors, including LSTM-based and transformer-based models, to assess whether failure patterns are model-specific or generalise across architectures. Cross-dataset evaluation on UCY [12] and Stanford Drone Dataset [13] scenes will further test generalisation. Integration of the failure rule layer into a closed-loop robot navigation system and evaluation of its safety impact in simulation will also be pursued.

6 Conclusion

I presented a model-agnostic framework for systematic failure mode analysis of pedestrian trajectory predictors. The framework combines input-space sensitivity analysis, risk mapping, and interpretable decision tree classifiers to identify and explain prediction failures without requiring access to model internals.

Applied to a Social Force Model on two scenes of the ETH pedestrian dataset, comprising 719 trajectories, the framework revealed that initial orientation is a consistent predictor of failure across scenes, while positional features exhibit scene-dependent importance. Decision tree classifiers achieved 79.3% and 82.0% accuracy on the ETH and Hotel scenes respectively, producing human-readable rules suitable for runtime safety monitoring.

The scene-dependent nature of failure patterns highlights the importance of scene-specific safety assessment and motivates the integration of adaptive safety monitors in robot navigation systems. The framework is publicly available at `github.com/pouyapd`.

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